

KNN Project - Bird Classifier

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Project source: <https://github.com/Podisti/KNN-bird-classifier/tree/main>

TASK DEFINITION

This project focuses on bird detection and classification. The goal is to be able to detect and classify a bird in a picture by using a convolutional neural network implemented using PyTorch and trained on a suitable dataset. Additionally, the network will make use of additional annotations of the birds' parts and their descriptions, and will learn to generate such descriptions for new bird images to be used alongside a dictionary to increase accuracy and justify the classification decision.

EXISTING SOLUTIONS

Bird classification is a challenging fine-grained image recognition task due to the high visual similarity between species and large variations in pose, lighting, and background. Recent advances in convolutional neural networks (CNNs) and transfer learning have significantly improved automated bird classification systems. A comprehensive overview of fine-grained image analysis methods is provided in Wei et al. (2019), which discusses the main challenges and deep learning approaches used for tasks such as bird species recognition.

Several studies have applied CNN-based architectures to bird classification. Dharaniya R et al. (2022) demonstrated the effectiveness of CNNs for extracting discriminative visual features directly from bird images. Cai (2023) used transfer learning with the EfficientNetB0 architecture and data augmentation to classify 525 bird species, achieving strong accuracy even with imbalanced datasets. More recent work by Jadhav et al. (2025) combines visual and acoustic information to improve classification robustness in wildlife monitoring applications.

DESCRIPTION OF OUR SOLUTION

Our solution is based on multi-task convolutional neural network, leveraging the ResNet50 architecture. Model consists of three heads, which simultaneously predict three outputs from an input image - the bird species, bounding box of the bird and the presence of specific birds' attributes (wings shape, tail length, primary color etc.). This approach creates synergy between those heads, where learning of bird attributes leads to more successful final classification of a bird's species alone.

We trained our model in a total of 15 epochs. Additionally, we utilized a technique called decreasing learning rate, where every 5 epochs, learning rate is divided by 10. That leads to more stable convergence and precise fine-tuning of model's weights.

Our innovation is implementing a hybrid classification mechanism that combines original visual prediction of the model and a statistic dictionary. This dictionary holds information about the most representative attributes for each species, calculated using mean aggregation of the training data. We applied TF-IDF (Term Frequency-Inverse Document Frequency) weighing to attributes. This penalizes very common attributes (such as "black eyes", "small size" etc.) and rewards unique attributes for each species. The final decision is based on smart logic tie-breaker with the following features:

- Dynamic Alpha - the logic module only activates if a confidence of the main visual model falls below a specific threshold. If the visual model is certain, logic module does not take a part in classification.
- Top-K Masking - to prevent model from hallucinating, the logic is restricted to considering only the top 3 candidates identified by the main visual model.
- Temperature Scaling - we used temperature-controlled Softmax to put more emphasis on attribute-based scores, ensuring a more confident selection.

Datasets

Two datasets were used while implementing this project. The first and the most necessary dataset is *Caltech-UCSD Birds-200-2011* (*CUB-200-2011*) by Krishna (2019). The dataset consists of 11788 bird images of 200 bird species. All data is annotated, these annotations include the birds' class, its bounding box, the location of its parts and binary

attributes for those parts. It also includes a train/test split. That means approximately 60 samples for each bird species, which might not be a sufficient amount to avoid overfitting and poor generalization. Therefore, we implemented several data augmentations to increase the amount of training samples for each species.

The second dataset is called *Intel Image Classification* by Bansal (2019), which contains around 25000 images of size 150x150 distributed under 6 categories - buildings, forest, glacier, mountain, sea and street. The dataset is used to expand the *CUB-200-2011* by one class (no bird), and allow for direct detection on top of classification.

Development environment

Implementation of our project was realized in Google Colab, using a Jupyter Notebook interface. This platform was chosen primarily for its access to cloud-based GPU acceleration using NVIDIA T4. That was crucial to train our model effectively in a "short" amount of time, compared to local training on an average personal computer. Google Colab also ensured very helpful integration with Google Drive and GitHub.

EXPERIMENTS

The evaluation was performed on the testing split of the extended CUB-200-2011 dataset. The experiments focused on both classification and localization performance of the proposed multi-task neural network.

The tested models were based on the ResNet18 and ResNet50 architectures. The evaluation measured Top-1 and Top-5 classification accuracy, mean Intersection over Union (IoU) for predicted bounding boxes, as well as IoU@0.50 and IoU@0.75 localization metrics. In addition, attribute prediction accuracy was evaluated for the predicted bird part attributes.

To better analyze the behavior of the models, per-class accuracy and normalized confusion matrices were generated. The experiments also identified the most frequently confused class pairs in order to highlight difficult fine-grained categories. Furthermore, softmax confidence scores were analyzed separately for correct and incorrect predictions, and confidence histograms were generated to evaluate model calibration and prediction reliability.

All evaluation results, including classification reports, confusion matrices, and confidence visualizations, were automatically exported into generated report files.

RESULTS

While reference model from original study reached an accuracy of 87 %, our main multi-task visual model consisting of 3 described heads reaches following accuracies:

- 80.7% Top-1 and 92.9% Top-5 accuracies on training dataset
- 81.3% Top-1 and 95.5% Top-5 accuracies on validation dataset
- 80.6% Top-1 and 95.9% Top-5 accuracies on testing dataset

Our additional hybrid mechanism is highly dependent on parameter tuning. In various test runs, we got results ranging from a slight decrease of 0.4 % to a 0.8 % increase in Top-1 accuracy. While the logic module successfully fixes errors where the visual model classifies the bird image incorrectly but correctly classifies its parts, it can sometimes override correct but low-confidence visual predictions.

The high Top-5 accuracy indicates that the correct species is usually included among the highest-confidence predictions, even in cases where the Top-1 prediction is incorrect. The attribute prediction branch achieved an accuracy of 92.4%, showing that the network successfully learned fine-grained visual bird characteristics.

Localization performance of the predicted bounding boxes reached a mean IoU of 0.63, with 78.6% of predictions exceeding IoU@0.50 and 25.4% exceeding IoU@0.75. These results suggest that the model is generally able to localize birds reliably, although highly precise localization remains challenging.

Confidence analysis further demonstrated that model confidence strongly correlates with prediction correctness. Correct predictions achieved an average confidence of 0.79, while incorrect predictions averaged only 0.40. This behavior indicates relatively good calibration of the classifier.

The confusion matrix and the most confused class pairs indicate that the model performs well on generic species recognition but struggles with subtle visual differences between closely related subspecies. The most common confusions occurred between visually similar bird species such as *Great Grey Shrike* and *Loggerhead Shrike*, *White-necked Raven* and *Common Raven*, or *Black-billed Cuckoo* and *Yellow-billed Cuckoo*.

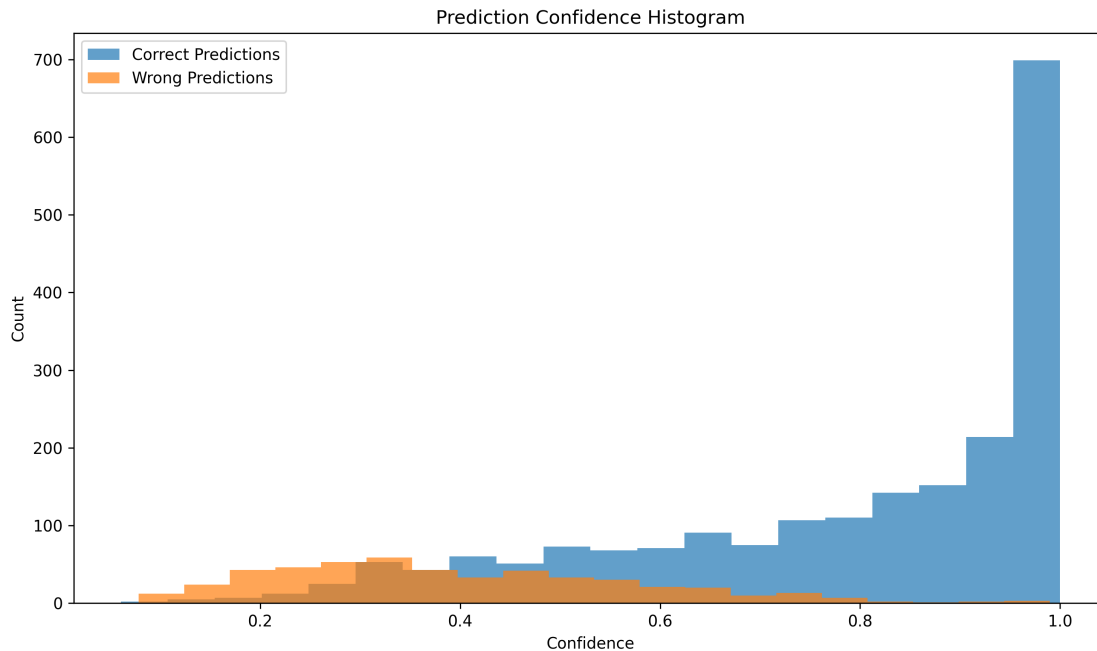


Figure 1. Distribution of confidence scores for correct and incorrect predictions.

CONCLUSION

The proposed multi-task neural network successfully fulfilled the main objective of detecting and classifying bird species from images while simultaneously predicting bounding boxes and visual attributes. The achieved Top-5 accuracy of nearly 96% shows that the correct species is usually present among the model's most confident predictions. The confidence diagram also shows the good calibration of the model.

The experiments also revealed the main limitation of the system. Most classification errors occurred between visually very similar species or subspecies, such as ravens, gulls, shrikes, and flycatchers, where even small visual differences are difficult to distinguish reliably. This behavior is reflected in the confusion matrix and corresponds to the challenging nature of fine-grained bird recognition tasks.

The localization branch achieved stable bounding box predictions with a mean IoU above 0.63, demonstrating that the model can reliably identify the bird region within an image. In addition, the attribute prediction branch achieved high accuracy and provided useful semantic information for the hybrid decision mechanism.

Overall, the project demonstrates that combining CNN-based visual classification with attribute-based reasoning creates a robust and interpretable bird recognition system capable of achieving strong performance on a challenging real-world dataset.

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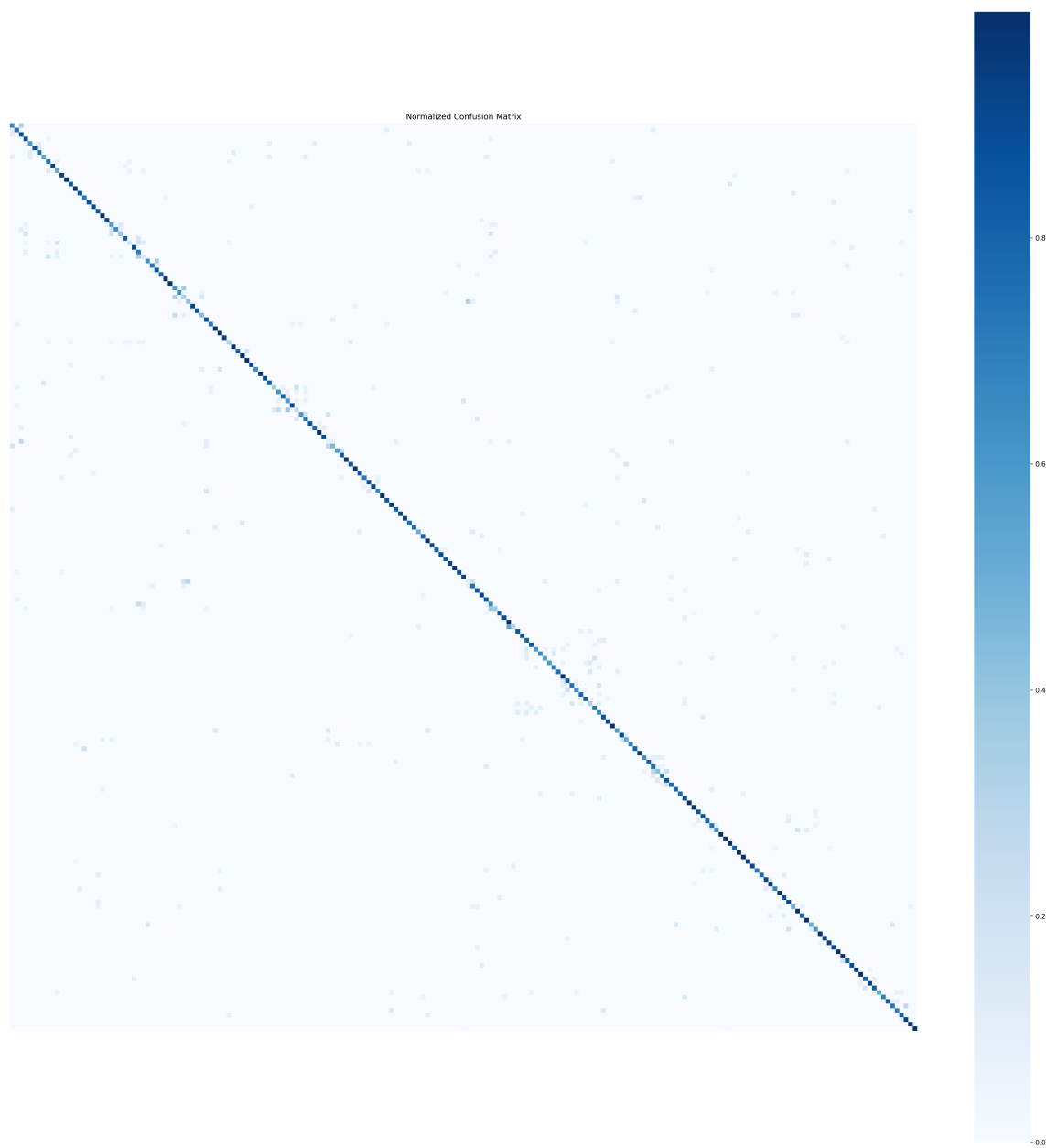


Figure 2. Normalized confusion matrix of the testing dataset predictions.